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Estimation of Time-varying OD Matrices for Metro Systems in Megacities

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Core problem and challenges

Core problem

- **Background**: In a multi-operator metro network, the OD matrix is essential for fair revenue distribution among operators, as passenger journeys often span multiple lines.
- **Data limitations**: The card-swiping system only records the entry and exit of the station and cannot obtain passengers' transfer path.
- **Current method**: The survey method is usually used, which is costly and has low accuracy.

How to accurately infer the time-varying OD matrix using only the total passenger volume at subway stations?

Technical challenges and solutions

- How to capture the time-varying characteristics of passenger flow?
  → Bayesian time-varying OD estimation model
- How to infer transfer behaviors?
  → Shortest travel time path assumption
- How to construct directional passenger flow constraints?
  → Weighted up/down decomposition of total flow

OD matrix (to be estimated)

Destination Sj (j=1,…,5)

Origin Si (i=1,…, 5)

	S1	S2	S3	S4	S5	Ti
S1	0	11	13	22	15	61
S2	6	0	17	6	13	42
S3	14	3	0	10	25	52
S4	13	12	19	0	20	64
S5	8	14	12	4	0	38
Tj	41	40	61	42	73	257

Boarding volume

Alighting volume

Observations

Current achievements and future work

Current achievements

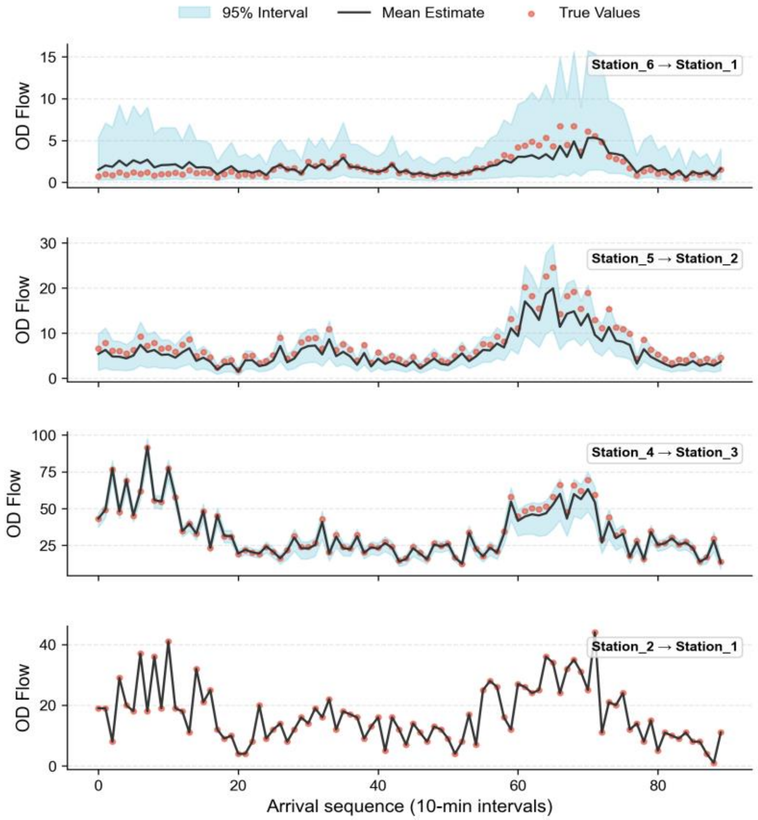

Figure 1. The method was verified through six-station simulation data

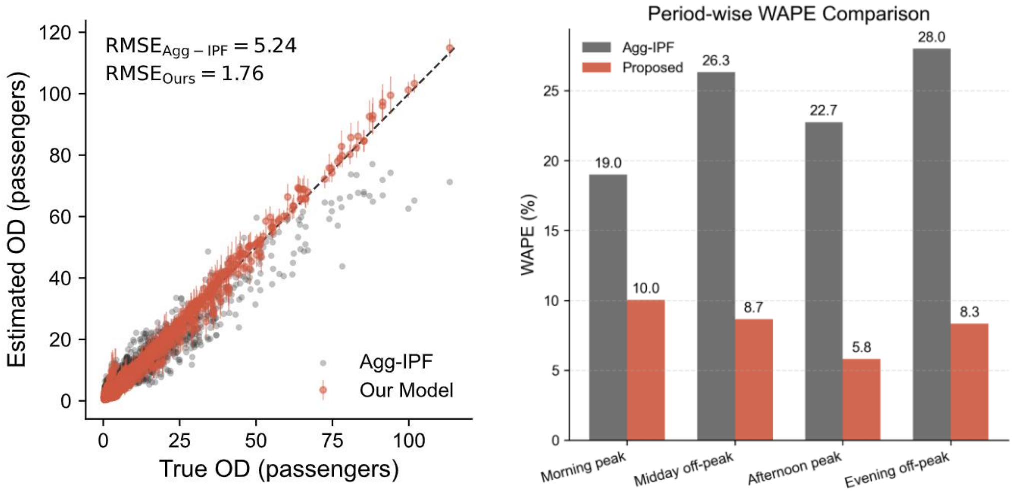

Figure 2. The method was compared with the traditional Agg-IPF method

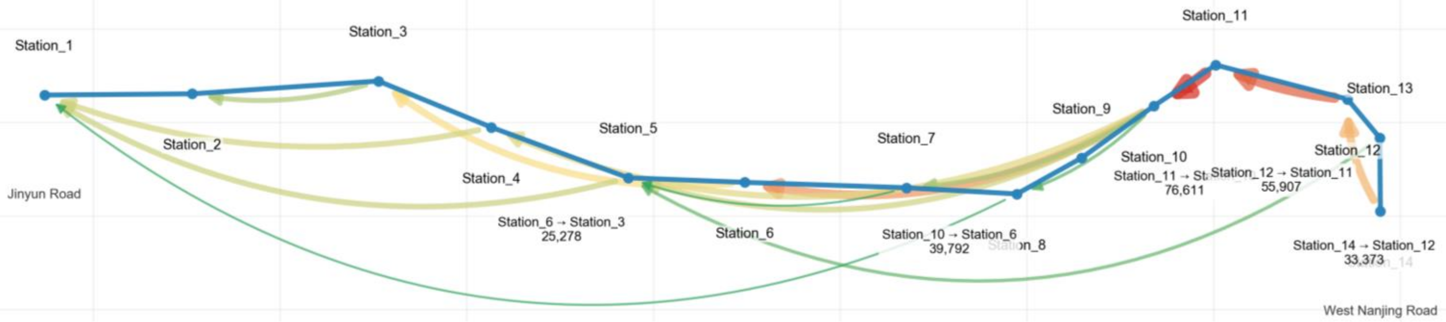

Figure 3. The method was applied to Shanghai Line 13

Future work

- Extend to more large-scale metro networks to test cross-city robustness and derive city-specific insights.
- Enable real-time OD estimation to support dynamic scheduling and emergency management.